

What is Health?

- Health is the state of complete physical, mental, and social well-being.
- Health increases productivity and ensures longevity.

Ways to Ensure Good Health

- Balanced diet
- Personal hygiene
- Exercise
- Awareness about prevention and control of diseases
- Proper waste disposal and control of vectors
- Vaccination

Why do Diseases Occur?

- Genetic reasons – Innate deficiencies and inheritable defects
- Infections
- Sedentary life style – Junk food, consumption of alcohols/drugs, lack of exercise

Pathogenic Diseases

Pathogens are the organisms that enter the human body through various means, then multiply, interfere with normal vital activities and causes harm to the human body. The diseases caused due to these pathogens are called pathogenic diseases.

Types of Pathogenic Diseases

1. Bacterial Disease

Typhoid

- Pathogen – *Salmonella typhi*
- Spreads through – Contaminated food and water
- Site of infection – Small intestine
- Symptoms – High fever, stomach pain, headache, loss of appetite, constipation, and intestinal perforations in severe cases
- Confirmatory test – **Widal test**

NOTE – A classic case in medicine, that of **Mary Mallon** nicknamed **Typhoid Mary**, is worth mentioning here. She was a cook by profession and was a typhoid carrier who continued to spread typhoid for several years through the food she prepared.

Pneumonia

- Pathogens – *Streptococcus pneumoniae* and *Haemophilus influenzae*
- Spreads through – Droplets/aerosols released from infected person, sharing of glasses or utensils
- Site of infection – Alveoli (gets filled with fluid, difficulty in breathing)
- Symptoms – Fever, chills, cough, headache, lips and nails become grey in severe cases

2. Viral Diseases

Common cold

- Pathogen – Rhino viruses
- Site of infection – Nose and respiratory passage
- Spreads through – Droplets released from coughing or sneezing, or contaminated objects
- Symptoms – Nasal congestion and discharge, sore throat, cough, headache, tiredness

3. Protozoan Diseases

Malaria

- Pathogen – *Plasmodium* sps. (*P. vivax*, *P. falciparum*, *P. malaria*)
- Vector – Female *Anopheles* mosquito
- Symptoms – High grade fever, chills

Amoebiasis

- Pathogen – *Entamoeba histolytica*
- Vector – Housefly
- Site of infection – Large intestine
- Symptoms – Constipation, abdominal pain, cramps, stools with mucous, and blood clots

4. Fungal Diseases

Ringworms

- Pathogens – Genera *Microsporum*, *Trichophyton*, and *Epidermophyton*
- Spreads through – Towels, clothes, comb (Fungus is acquired from soil)
- Symptoms – Appearance of dry, scaly lesions on various body parts with intense itching



Diagram showing ringworm affected area of the skin

5. Diseases Caused by Worms

Ascariasis

- Pathogen – Round worm, *Ascaris*
- Spreads through – Water, vegetables, fruits contaminated by faeces of infected person
- Symptoms – Internal bleeding, muscular pain, fever, anaemia, blockage of intestinal passage

Elephantiasis (filariasis)

- Pathogen – *Wuchereria* (*W.malayi* and *W.bancrofti*)
- Spreads through – Bite of female mosquito vector
- Symptom – Chronic inflammation of the organs, usually the lymphatic vessels of lower limb

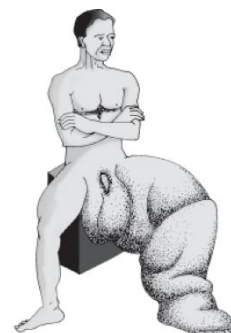
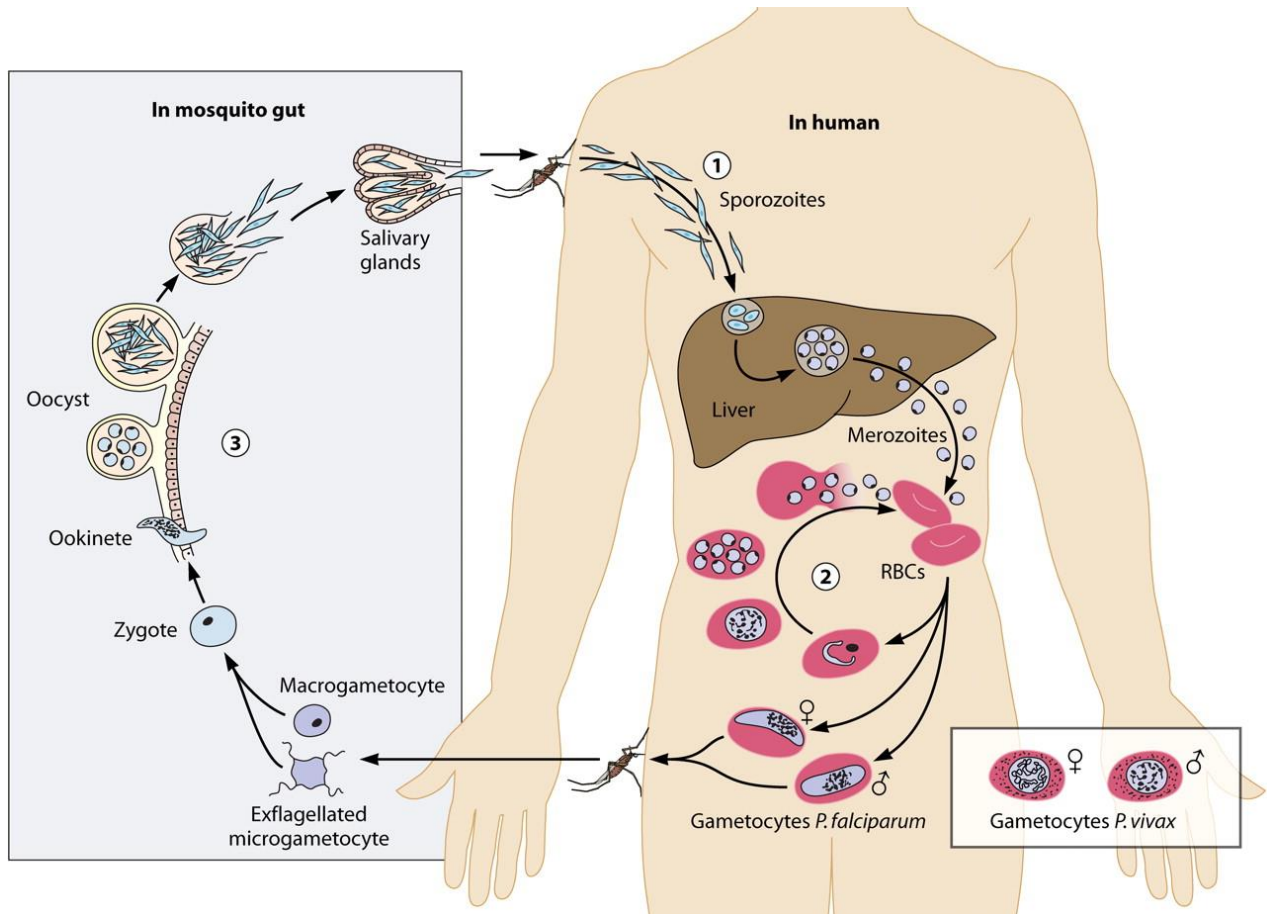


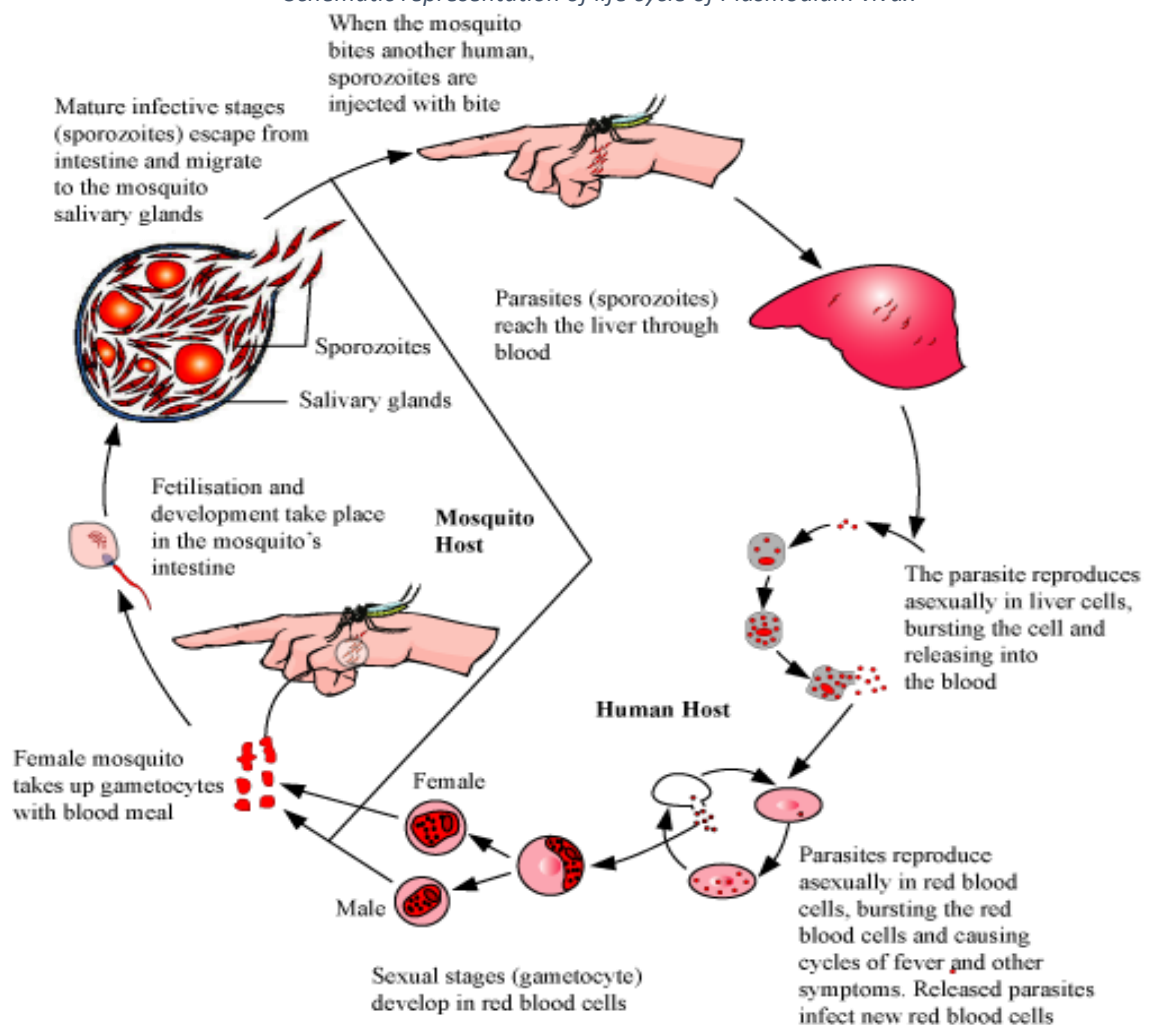
Diagram showing inflammation in one of the lower limbs due to elephantiasis

Life Cycle of *Plasmodium*

- *Plasmodium* requires two hosts to complete its life cycle.
- When female *Anopheles* mosquito bites a healthy human being, it releases *Plasmodium*, which lives in its body as sporozoite (infectious form).
- The parasites multiply (asexual reproduction) in the liver cells and finally burst the liver cells. Sporozoites are released in blood.
- Parasites enter RBCs and further multiply (asexual reproduction) here and finally burst RBCs also.
- Bursting of RBCs is accompanied by release of a toxic substance called **haemozoin** (associated with fever and chills).
- In the RBCs, only sporozoites change into gametocytes (sexual stage). Gametocytes multiply.
- When the diseased person is bitten by a female *Anopheles* mosquito, gametocytes are introduced into the mosquito.
- Gametocytes fertilise and develop inside the intestine of mosquito to form sporozoites.
- Sporozoites are stored in the salivary glands of mosquito and are released into the healthy person who is bitten by this mosquito.



Schematic representation of life cycle of *Plasmodium vivax*



Prevention and Control of Infectious Diseases

- Measures for personal hygiene include keeping the body clean; consumption of clean drinking water, food, vegetables, fruits, etc.
- Public hygiene includes proper disposal of waste and excreta; periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks and observing standard practices of hygiene in public catering. These measures are particularly essential where the infectious agents are transmitted through food and water such as typhoid, amoebiasis and ascariasis.
- In cases of air-borne diseases such as pneumonia and common cold, in addition to the above measures, close contact with the infected persons or their belongings should be avoided.
- For diseases such as malaria and filariasis that are transmitted through insect vectors, the most important measure is to control or eliminate the vectors and their breeding places. This can be achieved by avoiding stagnation of water in and around residential areas, regular cleaning of household coolers, use of mosquito nets, introducing fishes like *Gambusia* in ponds that feed on mosquito larvae, spraying of insecticides in ditches, drainage areas and swamps, etc. In addition, doors and windows should be provided with wire mesh to prevent the entry of mosquitoes. Such precautions have become all the more important especially in the light of recent widespread incidences of the vector-borne (*Aedes* mosquitoes) diseases like dengue and chikungunya in many parts of India.
- The use of vaccines and immunisation programmes have enabled us to completely eradicate a deadly disease like smallpox.
- A large number of other infectious diseases like polio, diphtheria, pneumonia and tetanus have been controlled to a large extent by the use of vaccines.

What is immunity?

- The ability of body to fight the disease-causing organisms is called immunity.

Types of immunity

- Immunity is of two types – innate immunity and acquired immunity.

1. Innate immunity – It is present from the time of birth. It is **non-specific**. It consists of 4 kinds of barriers.

- Physical barriers – Skin and mucus coating of respiratory, gastrointestinal, and urogenital tract prevent entry of microbes into body.
- Physiological barriers – Acid in stomach, saliva in mouth, tears from eyes
- Cellular barriers – Blood has leukocytes such as polymorpho nuclear leukocytes, monocytes, etc. and tissue has macrophages which phagocytose the microbes.
- Cytokine barriers – Special proteins called **interferons** are secreted by virus-infected cells that prevent the further spread of viral infection.

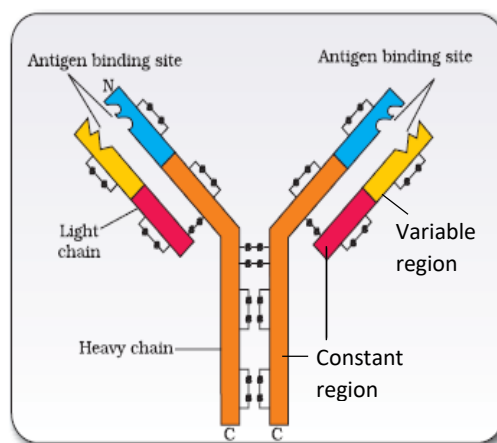
Local Defence System of the Body

Apart from these defence mechanisms, our body has some local defence systems, which act as the barrier and protect our bodies from local injuries and infections.

Some of these are given in the following table.

Organ	Function
Skin	- First line of defence; prevents entry of germs - Oil glands and sweat glands destroy the germs.
Nose	- Hair present in the nose filters the foreign particles. - These particles are thrown out during sneezing.
Eyes	Tear glands secrete tears containing lysozymes. Lysozyme prevents eye infections.
Mouth	Salivary enzymes kill the microbes.
HCl in stomach	Kills the microbes that reach the stomach
WBCs	Kill the microbes that enter the body through cuts and abrasions

2. **Acquired immunity** – It is acquired, which means that it is produced in response to an encounter with a pathogen based on memory. It is **pathogen specific**.
 - When a pathogen for the first time infects a person, low intensity immune response is generated called as **primary response**.
 - When the same pathogen attacks again, intensified immune response is generated called as **secondary response**, thereby preventing the occurrence of disease.
 - Acquired immunity involves two types of cells – B-lymphocytes and T- lymphocytes.
 - **B-lymphocytes** – Secrete proteins called **antibodies** in response to pathogens. **Antibodies** are specialized proteins with **4 peptide chains** (2 light and 2 heavy), hence denoted as **H₂L₂**. IgA IgM, IgE, IgG etc. are examples of some of the antibodies. They generate **humoral immune response** (found in blood plasma).



Structure of an antibody molecule

- **T-lymphocytes** – they help B-cells to produce antibodies. They generate **cell-mediated immune response**. This response helps the body to differentiate between 'self' and 'non-self' as occurs in case of **graft rejection**.

Acquired immunity is further of two types – Active and passive immunity.

1. **Active immunity** - resistance is developed by an individual due to a previous infection. This is the reason why a person has a very rare chance of getting a disease again, if he has suffered and survived from it once in his lifetime.

Two types of active immunity – Natural and artificial

Natural – Body gets immunity naturally when the person suffers from a disease. During a disease, the person produces antibodies to fight the infection, which remain in the body for a long time to prevent the occurrence of a disease on subsequent exposure to the same microbe. These **antibodies are produced by plasma cells** of the immune systems.

Artificial – It is imparted by vaccination.

2. **Passive immunity** is the acquired immunity that is provided to an individual by injecting the individual with serum containing antibodies from another individual or animal.

Two types of passive immunity – Natural and artificial

Natural – In this, readymade antibodies (**IgG**) from mother reach the foetus through placenta and the yellowish fluid **colostrum** (mother's first milk) secreted by mother during the initial days of lactation has abundant antibodies (**IgA**) to protect the infant.

Artificial – In this, serum antibodies from horse or other animals are isolated (they are first made stronger by repeatedly injecting the vaccination for the particular disease). These antibodies are injected into the humans to provide them immunity.

The serum containing antibodies against a particular antigen is called **antiserum**. For preparing the antiserum, the following steps are employed.

- i) First of all, the disease-causing microbe is cultured and toxin is separated from it.
- ii) The toxin is injected into the body of the animal in small amount.
- iii) After some days, stronger dose of toxin is given to the animal.
- iv) The immune system of the animal works against the toxin and secretes antibodies against it. Now the serum of the animal contains concentrate amount of antibodies against that particular disease.
- v) The blood is drawn from the animal and is allowed to coagulate in order to separate the serum.
- vi) This serum is collected in sterilised bottles.

Difference between Active and Passive immunity

Active immunity	Passive immunity
Body's own cells produce antibodies in response to an infection.	Antibodies from another person or animal are injected into the person.
Person's own immune response has to get activated and produce antibodies. Hence, this takes time.	Ready-made antibodies are injected in the person. Hence, it provides immediate cure.
This is harmless.	This may cause reactions.
It provides long lasting protection.	Its effects are short lived.

How does vaccination help?

- Vaccines are nothing but inactivated pathogens.
- These inactivated pathogens when introduced in the body produce a primary immune response and antibodies are produced against the pathogen.
- Memory B and T-cells are produced.
- Now when the pathogen again attacks the person, memory B and T-cells generate a massive immune response and the pathogen is killed.

NOTE –

- If a person is infected with some deadly microbes to which quick immune response is required as in **tetanus** (lock jaw) (bacterium – *Clostridium tetani*, we need to directly inject the preformed antibodies, or **antitoxin** (a preparation containing antibodies to the toxin).
- Even in cases of snakebites, the injection which is given to the patients, contain preformed antibodies against the snake venom. This type of immunisation is called **passive immunisation**.

Problems of immune system

Allergies

- Exaggerated immune response to certain antigens present in environment
- **Allergens** – Substances in response to which allergy is produced. E.g., animal dander, dust, pollen, etc.
- Antibodies involved – **IgE** type
- During allergic reactions, chemicals such as **histamines** (secreted by basophils and mast cells) and **serotonins** (import chemical and neurotransmitter in the human body. It is believed to help regulate mood and social behavior, appetite and digestion, sleep, memory and sexual desire and function)
- Symptoms – Sneezing, watery eyes, difficulty in breathing, etc.
- Allergy test – Patient is injected with small doses of allergens to monitor his response.
- **Antihistamines**, **adrenalins**, and **steroids** may be given so that the symptoms of allergy subside.

Autoimmunity

- In autoimmunity, body generates immune response against its own cells.
- Reasons – Genetic and other unknown reasons
- Example – **Rheumatoid arthritis** is an autoimmune disease (Rheumatoid arthritis is a chronic disease causing inflammation in the joints and resulting in painful deformity and immobility, especially in the fingers, wrists, feet and ankles)

Human immune system

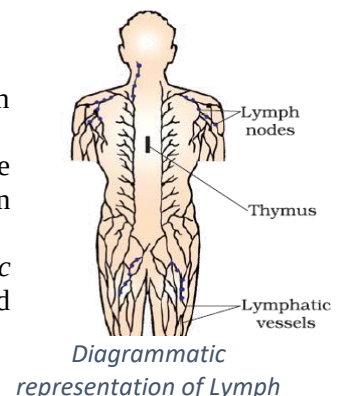
- Lymphoid organs are of two types – primary lymphoid organs and secondary lymphoid organs.

Primary lymphoid organs - consist of bone marrow and thymus. Here, immature lymphocytes are differentiated to form antigen-sensitive lymphocytes.

- **Bone marrow** – Here, all blood cells including lymphocytes are produced.
- **Thymus** – It is responsible for maturation of T-lymphocytes. This lobed organ is situated near the heart and keeps on reducing in size as the age increases.

Secondary lymphoid organs – Lymphocytes migrate here after attaining maturity. It includes spleen, lymph nodes tonsils, Peyer's patches, and appendix.

- **Spleen** – Large bean-shaped organ containing lymphocytes and phagocytes, which acts as a filter for blood. The main functions of the spleen are –
 - To produce immune response against blood borne antigens.
 - To remove particulate matter and aged blood cells mainly erythrocytes.
 - To produce blood cells during fetal life.
- **Lymph nodes** –
 - Each of a number of small swellings in the lymphatic system where lymph is filtered and lymphocytes are formed.
 - Located at different points throughout the immune system, they trap the antigens present in lymph or tissue fluid, and these antigens cause activation of lymphocytes and generation of immune response.
 - **Extra** - A *lymph node* is an oval- or kidney-shaped organ of the *lymphatic* system, present widely throughout the body including the armpit and stomach and linked by *lymphatic* vessels. *Lymph nodes* are major sites of B, T, and other immune cells.
- **MALT** (Mucosal-associated lymphoid tissue) – Lines major tracts (respiratory, digestive, urogenital, etc); constitutes **50% of lymphoid tissue** in body

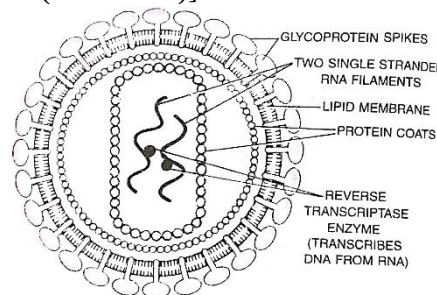


What Is Health?

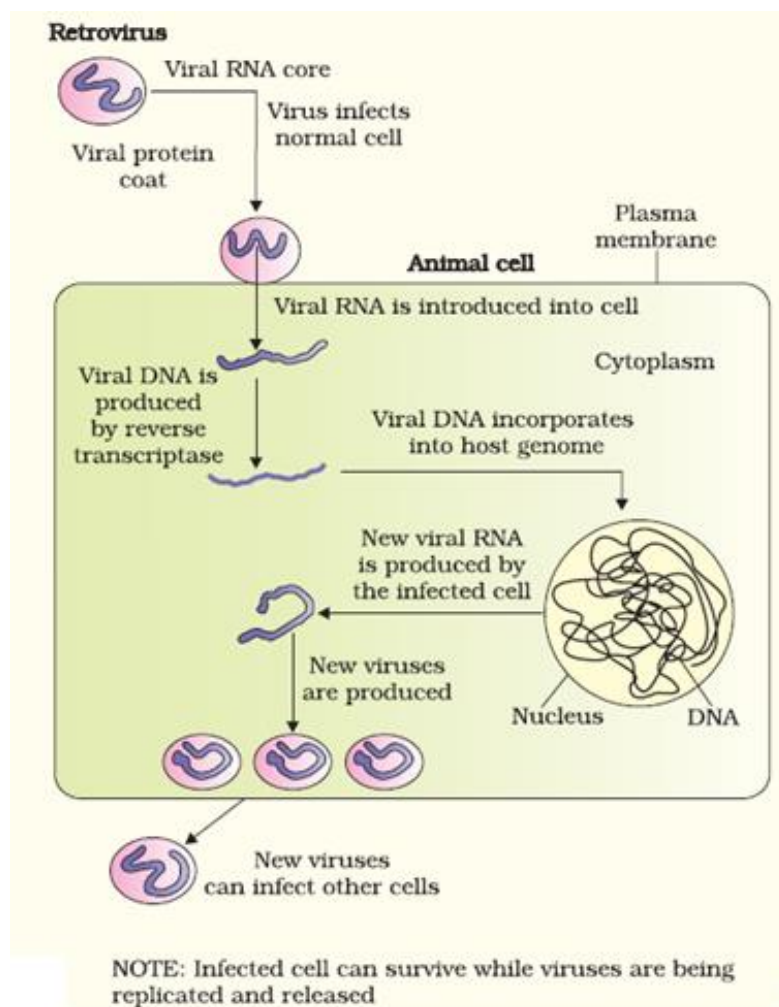
- According to WHO, health is defined **as the state of complete physical, mental, and social well-being.**
- Health does not merely mean freedom from diseases, but it covers the broader spectrum of the ability to perform all tasks up to one's full strength. It includes a state of psychological well-being where a person has positive attitude and motivation.
- **April 7** is celebrated as the **World Health Day.**
- Whenever we think about health, the term 'immune system' comes into our minds. It is the system of our body that keeps us healthy.

AIDS (Acquired Immuno Deficiency Syndrome)

- Caused by **HIV** (Human Immunodeficiency Virus) [HIV is a retrovirus (RNA virus)]
- Transmission of HIV occurs through:**
 - Sexual contact with infected person
 - Sharing infected needles (as in case of intravenous drug abusers)
 - Transfusion of contaminated blood
 - Infected mother to child through placenta
- Time lag between infection and appearance of symptoms – Few months to many years (5-10 years)



How does AIDS infection spread? (Replication of retrovirus – HIV)



- Virus enters the body of a person and enters macrophages.
- Here, virus replicates (viral RNA reverse transcribes to viral DNA, which gets incorporated into hosts DNA and subsequently new viral particles are produced).
- Macrophages become a virtual HIV factory.
- Thereafter, HIV enters helper T-lymphocytes, replicates, and produces progenies.
- As the progenies are released, they attack other T-lymphocytes.
- Therefore, T-lymphocytes start decreasing in number and immune response of the person becomes weak.
- Later, serious infections with Mycoplasma, Viruses, Fungi and even parasites like Toxoplasma occur and person suffers horrible pain and eventually dies a terrible death.

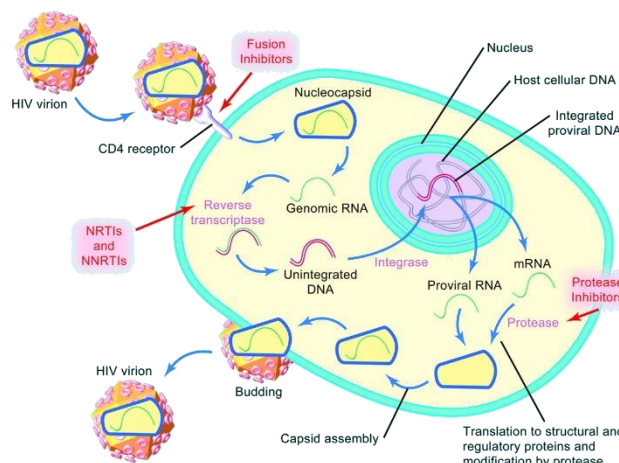
Diagnosis of AIDS – By **ELISA** (Enzyme Linked Immuno Sorbent Assay) and **Western blotting** test. Western blotting test is employed for confirmation of ELISA positive cases.

Treatment – No permanent cure; Antiretroviral therapies (ART) can prolong the life of patient

Extra information

- AIDS can be partially treated by Antiretroviral drugs, which can be used to control the retroviral replication at different levels

1. Fusion / Entry inhibitors
2. Reverse transcriptase inhibitors
3. Integrase inhibitors
4. Protease inhibitors



- AZT** (azidothymidine/zidovudine), drug used to delay development of AIDS.
- AZT belongs to a group of drugs known as **nucleoside reverse transcriptase inhibitors** (NRTIs).
- In **1987** AZT became the first of these drugs to be approved by the **U.S. Food and Drug Administration** for the purpose of prolonging the lives of AIDS patients

Prevention of AIDS

- Ensuring use of disposable syringes
- Screening blood from blood banks
- Advocating safe sex
- NACO** (National AIDS Control Organization) and many NGOs are doing a lot to create awareness among people.

Cancer

- The process of development of cancer is called **oncogenic transformation**.
- Normal cells have the property of contact inhibition** (stoppage of growth on coming in contact with other cells), but **cancer cells lose contact inhibition property**.
- As a result, cancer cells divide continuously to give rise to mass of cells (tumours).
- Tumours are of 2 types – benign and malignant.
- Benign tumours** – Remain confined to their original location and do not spread
- Malignant tumours**– These exhibit **metastasis** i.e., the cells sloughed from such tumours reach distant sites and wherever they reach, new tumour is formed.
- Malignant tumours actually represent cancer. The cells actively divide, grow, and starve the normal cells of vital nutrients.

Causes of cancer

- Carcinogens** – Physical, chemical, and biological agents that cause cancer Example - ionizing radiations (X-rays and gamma rays), non-ionizing radiations (UV)
- Oncogenic (cancer-causing) viruses** – they have viral **oncogenes** (cancer-causing genes).
- Sometimes normal genes in our body called **proto-oncogenes** get converted into **cellular oncogenes** (c-onc) that cause cancer.

Diagnosing cancer

- Biopsy and histopathological studies
- Biopsy** – Suspected tissue is cut into thin sections and examined microscopically
- Radiography - CT scan** (computed tomography), and **MRI** (Magnetic resonance imaging) are techniques of diagnosing cancers.

- **C T Scan** – 3-D imaging of internals of an organ is generated by X-rays.
- **MRI Scan** – Pathological and physiological changes in a living tissue are detected by using magnetic fields and non-ionising radiations.
- Immunological and molecular biological diagnostic techniques can all be used to detect cancers.
- Identifying certain genes, which make an individual susceptible to cancers, can help to prevent cancers.

Treatment of cancer

- **Radiotherapy** – Tumour cells are irradiated to death. Also, proper care is taken for protecting surrounding normal tissues.
- **Chemotherapy** – Drugs specific for particular tumours are used to kill cancer cells. They have side effects such as hair loss, anaemia, etc.
- **Immunotherapy** – Biological response modifiers such as α -interferons are used. They activate the immune system of patient and helps in destroying the tumour.

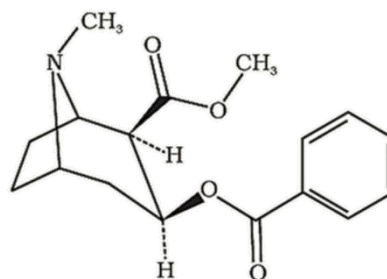
Commonly Abused Drugs

Opioids (Heroin/smack)

- Source: Acetylation of morphine extracted from the latex of poppy plants (*Papaver somniferum*)
- Consumed by: Snorting or injection
- Properties: White, bitter and odourless
- Mode of action: Binds to opioid receptors present in the CNS and GI tract
- Effect: It is a depressant; slows down body functions, pain killer



Opium poppy



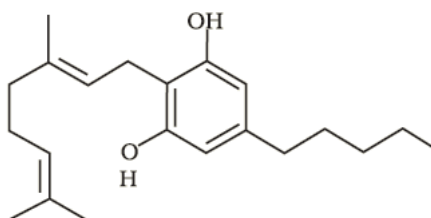
Chemical structure of Morphine

Cannabinoids

- Source: Inflorescences of the plant *Cannabis sativa*
- Consumed by: Inhalation or oral ingestion
- Mode of action: Binds to cannabinoid receptors present in the brain
- Effect: Affects the cardiovascular system



Leaves of Cannabis sativa



Skeletal structure of cannabinoid molecule

- Combination of flower tops, leaves and the resin of Cannabis plant are used to produce **marijuana**, **hashish**, **charas** and **ganja**.
- These days **cannabinoids** are also being abused by some sportspersons.

Coca alkaloids or Cocaine (coke / crack)

- Source: Coca plant *Erythroxylum coca*, found in South America
- Commonly called as **coke** or **crack**
- Consumed by: Snorting
- Mode of action: Interference with transfer of neurotransmitter, dopamine
- Effect: Stimulates the CNS, producing a sense of euphoria and increased energy; excessive dosages cause **hallucination**
- Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*.



Flowering branch of *Datura*

Drugs Normally Used as Medicines

- Drugs like barbiturates, amphetamines, benzodiazepines, LSD (Lysergic acid diethyl amides) are used as medicines to help patients with mental illness and insomnia.
- Morphine: It is a pain killer which is used for patients who have undergone surgery, but it is also abused.

Nicotine

- Present in tobacco, which is smoked, chewed or snuffed
- Mode of action: Stimulates the adrenal gland to release adrenaline and nor-adrenaline
- Effect: Increases blood pressure and heart rate.

III Effects of Smoking

- Increased risk of diseases like bronchitis, emphysema, coronary heart disease, gastric ulcer and cancer (throat, lung and urinary bladder)
- Increased carbon monoxide levels in blood, leading to oxygen deficiency

Alcohol / Drug Abuse

Causes of alcohol/ Drug Abuse

- Alcohol / drug abuse normally starts in adolescence (period between 12-18 yrs – transition phase between childhood and adulthood).
- Many adolescents are motivated towards drugs/ alcohol due to curiosity and experimentation.
- Peer pressure, academic stress, unstable family structure further incline youth towards alcohol/ drug abuse.
- Perception of consuming alcohol / drug being cool and progressive and use of alcohol/drug in television, movies, etc. further promote this habit.

Alcohol/ Drug Addiction

- When a person uses alcohol/ drug repeatedly, he becomes addicted.
- Addiction refers to psychological attachment to certain effects such as euphoria and temporary feeling of well-being associated with use of alcohol or drugs.
- In addiction, tolerance level of receptors present in our body increases towards the drug.
- This drives the person to use them even when they are not required or when they tend to harm his health / family life.
- Subsequently, the user runs into a vicious circle of addiction and subsequent dependence.
- Dependence leads to manifestation of withdrawal syndrome on discontinuation of use.
- **Withdrawal syndrome** – Anxiety, nausea, sweating, shakiness, and sometimes may be lethal

Effects of Alcohol/ Drug Abuse

- Immediate effect – Vandalism, violence, and reckless behaviour
- Drop in academic performance, lack of interest in personal hygiene, rebellious behaviour, and change in eating and sleeping patterns, weight and appetite fluctuations
- Mental, psychological, and financial loss not only to the user, but also to his family
- Those who take drugs intravenously have a high risk of acquiring deadly diseases such as AIDS and hepatitis B.
- Damage to nervous system and liver (cirrhosis)

- **Use of anabolic steroids by sports person have adverse effects:**
 - **In females** – Increase of masculinity, aggressiveness, depression, abnormal menstrual cycle, facial hair growth, enlargement of clitoris, and deepening of voice
 - **In males** – Acne, aggressiveness, depression, reduction in size of testicles, decreased sperm production, enlargement of prostate gland, breast enlargement, premature baldness
- Ultimately, prolonged use of alcohol/drugs leads to coma and death.

Preventing Alcohol/ Drug Abuse

- It is better to prevent the inclination of an individual towards alcohol/ drugs right from adolescence. Some of the ways of prevention are:
 - **Avoid peer pressure** – understand the unique personality and capabilities of a child
 - **Education and counseling** – a child must be taught to accept success and failure equally. Especially during adolescence, he must be inclined towards constructive activities such as music, yoga, sports, reading based on his interest.
 - **Help from parents and peers** – this includes proper guidance, advice, and trust to overcome problems such as stress and guilt.
 - **Identifying danger signals** – if any sign of symptom of alcohol / drug abuse is seen in the adolescent by family or friends, then it should not be ignored because prevention is better than cure.
 - **Seeking medical help** – Psychologists and rehabilitation programs surely help an addict. Medical help should be sought to prevent further damage.

SUMMARY

Health is not just the absence of disease. It is a state of complete physical, mental, social and psychological well-being. Diseases like typhoid, cholera, pneumonia, fungal infections of skin, malaria and many others are a major cause of distress to human beings. Vector-borne diseases like malaria especially one caused by *Plasmodium falciparum*, if not treated, may prove fatal. Besides personal cleanliness and hygiene, public health measures like proper disposal of waste, decontamination of drinking water, control of vectors like mosquitoes and immunisation are very helpful in preventing these diseases. Our immune system plays the major role in preventing these diseases when we are exposed to disease-causing agents. The innate defences of our body like skin, mucous membranes, antimicrobial substances present in our tears, saliva and the phagocytic cells help to block the entry of pathogens into our body. If the pathogens succeed in gaining entry to our body, specific antibodies (humoral immune response) and cells (cell mediated immune response) serve to kill these pathogens. Immune system has memory. On subsequent exposure to same pathogen, the immune response is rapid and more intense. This forms the basis of protection afforded by vaccination and immunisation. Among other diseases, AIDS and cancer kill a large number of individuals worldwide. AIDS caused by the human immunodeficiency virus (HIV) is fatal but can be prevented if certain precautions are taken. Many cancers are curable if detected early and appropriate therapeutic measures are taken. Of late, drug and alcohol abuse among youth and adolescents is becoming another cause of concern. Because of the addictive nature of alcohol and drugs, and their perceived benefits like relief from stress, a person may try taking these in the face of peer pressure, examinations-related and competition-related stresses. In doing so, he/she may get addicted to them. Education about their harmful effects, counselling and seeking immediate professional and medical help would totally relieve the individual from these evils.